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Department of Computer Engineering

Course Network Management **Tutorials** Lecturer: Dr S. C. Forbacha

Section A: MCQ

1) An approach which is reliant on a string of tracer-route like probe packets that would enable it to build a network map is called: 1 mark
   (a) SNMP-based approaches
   (b) Route analytics approaches
   (c) Active probing based approach
   (d) None of the above
2) A type of network in which all the nodes on a segment are connected to the same physical cable is called: 1 mark
   (a) Ring network
   (b) Mesh network
   (c) Bus network
   (d) Tree network
3) A network protocol that takes an IP address in an IP datagram and translates it into the appropriate medium access control layer address for delivery on a LAN is referred to as: 1 mark
   (a) Internet Protocol
   (b) Transmission Control Protocol
   (c) Network Address Translation
   (d) Address Resolution Protocol
4) A transport protocol that does not establish connections plus does not attempt to keep data packets in sequence as well as not watching those that have existed for too long is known as: 1 mark
   (a) Transmission Control Protocol
   (b) User Datagram Protocol
   (c) Internet Control Message Protocol
   (d) Dynamic Configuration Protocol
5) Computers that store network software and shared or private user files are known as: 1 mark
   (a) Servers
   (b) Workstations
   (c) Mainframes
   (d) Routers
6) The most popular type of cabling is: 1 mark
   (a) The unshielded twisted-pair
   (b) The shielded twisted-pair
   (c) The coaxial
   (d) The optic fibre
7) The measure of how fast we can actually send data through a network is referred to as: 1 mark
   (a) Latency
   (b) Propagation
   (c) Throughput
   (d) Transmission

Section B

1) State the two advantages of having protocol standards. 4 marks
2) Differentiate between a physical and a logical topology. 2 marks
3) Explain the method of operation of star topology giving its advantages and disadvantages. 8 marks
4) What is the difference between a hub and a switch? 2 marks
5) What is a network management system? 2 marks
6) List and briefly define the key elements of SNMP. 8 marks
7) What are the differences among SNMPv1, SNMPv2, and SNMPv3?
8) When a router has the IP datagram, it may make several decisions affecting the datagram's future. State and briefly explain these 3 functions. 6 marks
9) Because SNMP uses two different port numbers (UDP ports 161 and 162), a single system can easily run both a manager and an agent. What would happen if the same port number were used for both? 4 marks
10) What is the difference between a manager and an agent in SNMP? 2 marks
11) How can Remote Network Monitoring be used to assist SNMP? 5 marks
12) What is the function of the Simple Network Management Protocol? 4 marks
13) What should be found in the control center for a network operation? 6 marks
14) List the three most important skills a network administrator should possess. 6 marks
15) Differentiate between a network operation system and a standard operation system. Please support your answer with the 3 functions that are carried out by each. 8 marks
16) Name and briefly explain 3 approaches that are deployed in network mapping. 6 marks
17) Discuss the concept Information Management Base. 6 marks
18) Calculate the throughput of a network with a bandwidth of 10 Mbps that can pass only an average of 16,000 frames per minute with each frame carrying an average of 10,000 bits. 2 marks
19) What are the propagation time and the transmission time for a 20-Mbyte message (an image) if the bandwidth of the network is 1 Mbps? Assume that the distance between the sender and the receiver is 12,000 km and that light travels at 2.4×10^8 m/s. 4 marks (2,2)